Day-19

DTP/VTP (No

DTP (Dynamic Trunking Protocol) & VTP (VLAN Trunking Protocol).
They are CISCO proprietary, they run only on CISCO devices.

NOTE: DTP & VTP were removed from CCNA exam topics list for new CCNA (200-301) but we need to learn as we may still get some questions.

DTP (Dynamic Trunking Protocol)

- DTP is a CISCO proprietary protocol that allows CISCO switches to dynamically determine their interface status (access or trunk) without manual configuration.
- DTP is enabled by default on all Cisco switch interfaces.
- So, far, we have been manually configuring switchports using these commands:
  switchport mode access
  OR
  switchport mode trunk.
- For security purposes, manual config is recommend. DTP should be disabled on all switch ports.

Configuring through DTP.

i) switchport mode dynamic ?
   a) auto: set negotiation parameter to AUTO
   b) desirable: Set negotiation parameter to DESIREABLE

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Switchport mode dynamic ~~desirable~~ desirable

A switchport in "dynamic desirable" mode will actively try to form a trunk with other Cisco switches. It will form a trunk if connected to other switchport in following modes.

switchport mode trunk
switchport mode dynamic desirable.
switchport mode dynamic auto.

- We can see this by command

SW1# show int g0/0 switchport
Administrative: dynamic desirable
Operational Mode: trunk.

- If the other switch has "switchport mode access" it will form access not tru

Switchport mode dynamic auto

A switchport in ~~trunk~~ dynamic auto mode won't actively try to form a trunk with other Cisco switches, however it will form a trunk if the switch connected to it is actively trying to form a trunk. It will form trunk with a switchport in following cases:

- switchport mode trunk
- switchport mode dynamic desirable



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Administrative Mode	Trunk	Dynamic Desirable	Access	Dynamic Auto
Trunk	Trunk	Trunk	X	Trunk
Dynamic Desirable	Trunk	Trunk	Access	Trunk
Access	X	Access	Access	Access
Dynamic Auto	Trunk	Trunk	Access	Access

DTP won't form a trunk with a router, PC, etc. The switchport will be in access mode.

- On older switches, "switchport mode dynamic desirable" was the default.
- On newer, "switchport mode dynamic auto" is default administrative mode
- We can disable DTP negotiation on an ~~interable~~ interface with the command, switchport nonegotiate.

- Configuring an access port with switchport mode access also disables DTP negotiation on interface.

Trunk encapsulation via DTP negotiation

- Switches that support 802.1Q & ISL ~~can~~ trunk encapsulations can use DTP to negotiate the encapsulation ~~to~~ they'll use.
- The negotiation is enabled by default, as the default trunk encapsulation mode is switchport trunk encapsulation negotiate
- ISL is favoured over 802.1Q, so if both switches support ISL it'll be selected.
- DTP frames are sent in VLAN1 when using ISL, or in the native VLAN when using 802.1Q (the default native VLAN is VLAN1 however).

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VTP (VLAN Trunking Protocol)

- VTP allows us to configure VLANs on a central VTP server switch, and the other switches (VTP Clients) will synchronize their VLAN database to the server.
- It is designed for large networks with many VLANS, so that you don't have to configure each VLAN on every switch.
- It is rarely used & is recommended not to use.
- There are 3 versions: 1, 2 & 3
- There are 3 VTP modes: server, client & transparent
- Cisco switches operate in VTP server mode by default.

Let's see 3 VTP modes.

i) VTP Servers.

- Can add/modify/delete VLANS
- Store the VLAN database in non-volatile RAM (NVRAM) (so data won't be erased when
- Will increase revision no. every time a VLAN is added/mod deleted.
- Will advertise the latest version of VLAN database on trunk interface & VTP clients will synchronize their VLAN database to it
- VTP servers also functions as a VTP client.
- Therefore a VTP server will synchronize to other VTP server with a higher revision number.

ii) VTP Clients

- Cannot add/modify/delete VLANs.
- Do not store the VLAN database in NVRAM (in VTPv3, they do)
- Will synchronize their VLAN database to the server with the highest revision no. in their VTP do.
- Will advertise their VLAN database, and forward VTP advertisement to other clients over their trunk ports.

iii) VTP transparent.

- Doesn't participate in the VTP domain (doesn't sync in VLAN database)
- Maintains its own VLAN database in NVRAM. If can add/modify/delete VLANs, but they won't be advertised to other switches.
- Will forward VTP advertisements that are in the same domain as it.

Some Commands

show vtp status
(VTP v1/2 don't support extended VLAN range (1006-4094). Only VTP VTPv3 supports them)

vtp domain cisco
(If we add/delete/modify a VLAN then revision no. increases by 1)

- If a switch with no VTP domain (domain = NULL) receives a VTP advertisment with a VTP domain name, it will auto join that VTP domain
- If a switch receives VTP advertisement in same VTP domain, with higher revision no, it will update its VLAN database to match.

One danger of VTP:

If you connect an old switch with higher revision no. and to your network (and VTP domain name matches) all switches in the domain will sync their VLAN database to that switch

Some command (ctd----)

- vtp mode client

(client mode में vlan बनाने error aayega)

Changing VTP domain to an unused domain will reset the revision no. to 0.

Changing VTP mode to transparent will also reset revision no. to 0.

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